CVEs:2025-29927-Next.js Middleware Authentication Bypass –

Overview

This task focused on identifying a modern web stack and understanding how attackers use stack fingerprinting to discover and exploit known vulnerabilities.

The target application was built using Next.js and was vulnerable to:

CVE-2025-29927

This vulnerability allowed attackers to bypass authentication and access protected pages without logging in.

What is Next.js?

Next.js is a React-based web framework used for:

- dashboards,
- SaaS applications,
- admin panels,
- and modern websites.

It runs on:

- Node.js
- React
- server-side rendering technologies.

What Technique Was Used?

Stack Fingerprinting

This means identifying the technology running behind the website using passive signals like:

- HTTP headers,
- cookies,
- error messages,
- and URL patterns

How the Stack Was Identified

The following command was used:

curl -I http://TARGET:3001/
The response showed:

Other indicators:

- /_next/static/
- window.__next_f

These confirmed the application was running Next.js with the App Router architecture

What Component Was Vulnerable?

Middleware

Next.js uses a feature called:middleware.

Middleware works like a security checkpoint before users access protected pages.

Example:

- user requests /dashboard
- middleware checks login session
- if not logged in → redirects to /login

What Caused the Vulnerability?

Next.js internally used a special header:

x-middleware-subrequest
This header was supposed to be used only by the framework internally.

Its purpose was:

- preventing middleware loops,
- and managing internal requests.

The Main Problem

The framework trusted this internal header even when it came from external users.

There was:

- no validation,
- no verification,
- and no check whether the request was internal.

Because of this, attackers could manually add the header themselves.

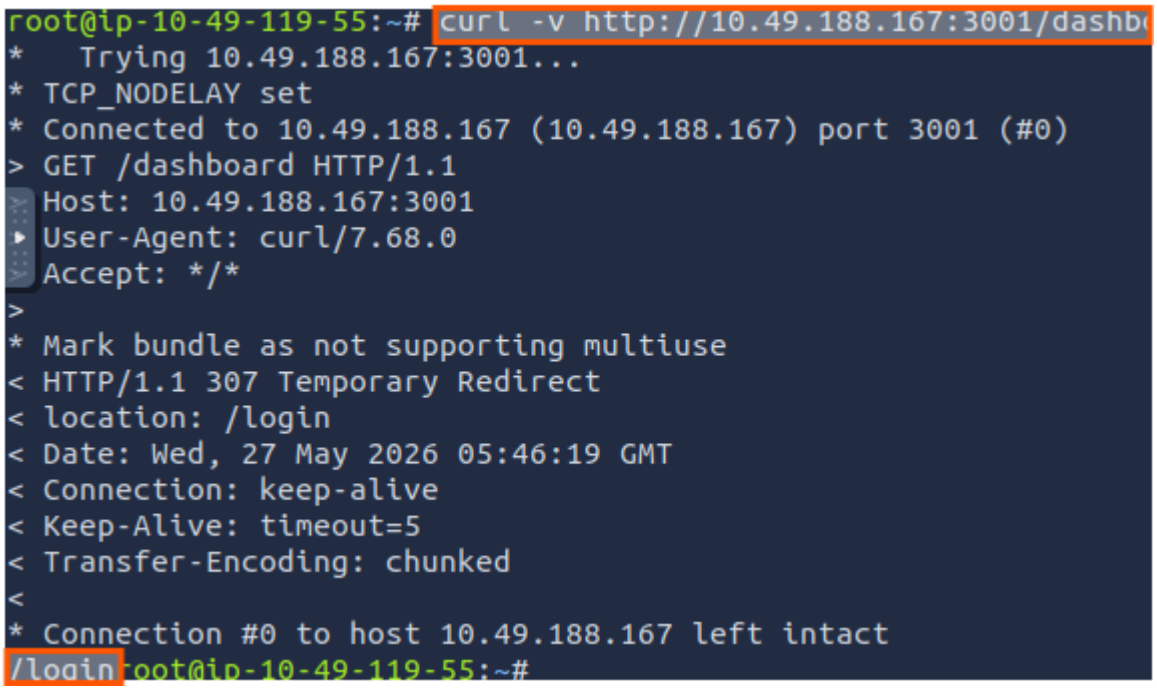
How the Exploit Worked

Normally:

- visiting /dashboard
- redirects unauthenticated users to /login.

Example:

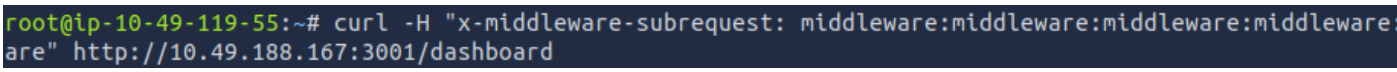
curl -v http://TARGET:3001/dashboard
Response:



This showed middleware protection was active.

Exploitation Technique

The attacker manually added the internal middleware header:



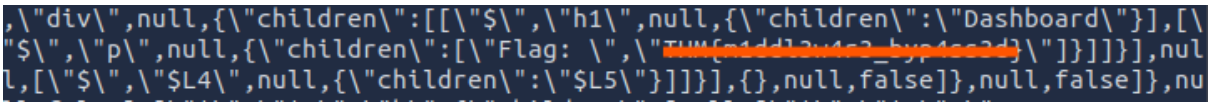
What Happened Internally

When Next.js received the request:

- it detected the special header,
- assumed middleware already executed,
- skipped authentication checks,
- and directly loaded the protected page.

Result

The attacker accessed the protected dashboard without login



This is called:

authentication bypass.

Why This Vulnerability Was Dangerous

This vulnerability was critical because:

- no credentials were needed,
- no brute force was required,
- exploitation required only one HTTP header,
- and many production applications relied entirely on middleware authentication.

Root Cause

The root cause was:

- trusting internal framework headers from external users.
- The application failed to properly validate internal requests.

Key Learning

This task demonstrated:

- how stack fingerprinting helps attackers,
- how framework-specific vulnerabilities are identified,
- and why modern web frameworks can create new attack surfaces.

It also showed the importance of:

- validating internal headers,
- secure middleware design,
- and proper authentication checks.

Mitigation

Recommended fixes:

- update Next.js to patched versions,
- validate internal headers,
- avoid relying only on middleware for authorization,
- implement backend access control checks.

Reference

- nvd.nist.gov/vuln/detail/CVE-2025-29927
- TryHackMe.com